



ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

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GAMIFIED LEARNING PLATFORM WITH PERFORMANCE ANALYTICS FOR FISH FARMING SKILLS TRAINING IN ZAMBIA

*A Project Proposal submitted to Zambia University College of Technology as a partial fulfilment
for the award of a Bachelor of Information Technology*

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CHAPTER 1: INTRODUCTION

1.1 Introduction and Project Background

Fish is a critical source of protein and income in Zambia, yet national demand consistently outstrips domestic supply. Aquaculture, the farming of fish in managed ponds, cages, and tanks, has become one of the fastest-growing agricultural sub-sectors in Zambia and across Sub-Saharan Africa, and is widely promoted by government and development partners as a route to food security, rural employment, and household income. The sector's continued growth, however, is constrained by a persistent skills gap. Most small-scale and emerging fish farmers begin with little or no formal training in pond construction, water-quality management, feeding, or disease control, and learn largely by trial and error. This approach frequently results in poor yields, avoidable stock losses, and farm abandonment.

Agricultural extension, the main channel through which farmers traditionally receive technical training, is severely stretched. Extension officers cover large geographic areas with limited transport and budgets, reach individual farmers infrequently, and cannot sustain the repeated, structured instruction that practical skill-building requires. Where dedicated training programmes do exist, they are often one-off workshops with no follow-up, no means of keeping farmers engaged between sessions, and no mechanism for trainers to see which farmers have actually mastered which skills.

Gamification, the application of game-design elements such as points, badges, leaderboards, experience points (XP), and achievement milestones to non-game contexts, has been shown in a substantial body of international research to significantly improve motivation, engagement, and completion when applied thoughtfully to learning. Applied to fish-farming training, gamification can transform a static set of lessons into a structured, rewarding progression that encourages farmers to return and complete each competency. Visible progress indicators and peer leaderboards supply the social motivation that sustains effort over time.

Equally important is trainer visibility. In a digital training context, an extension officer cannot observe a farmer's pond, watch a technique being applied, or read informal cues of confusion. Without a performance analytics dashboard that aggregates each learner's module completion, quiz scores, and engagement, trainers are effectively operating blind. They cannot easily identify

which farmers are struggling, which topics cause the most difficulty, or which learners are ready to apply their skills in the field.

This project proposes the design, development, and evaluation of a Gamified Learning Platform with Performance Analytics for fish-farming skills training, deployed at a single Zambian site such as an agricultural training institution, a fish-farming cooperative, or an extension programme, as a pilot study. The platform integrates a gamification engine that awards points, badges, and leaderboard rankings for completing aquaculture training activities, together with a real-time analytics dashboard that gives trainers individual and cohort-level performance metrics. It is built on a React.js and Node.js full-stack architecture with a MySQL database, designed to tolerate the low-bandwidth, intermittently connected conditions typical of rural farming areas, and evaluated through pre- and post-deployment learner surveys and platform usage analysis.

1.2 Problem Statement

Fish-farming training in Zambia currently faces four interconnected problems that undermine both learner engagement and trainer effectiveness. First, training is delivered passively, through occasional workshops, printed manuals, or unstructured advice, with no reward or recognition mechanism. This leaves farmers with low motivation to complete a full learning programme and no visible measure of their progress beyond eventual success or failure at the pond. Second, the absence of a real-time performance analytics dashboard means trainers and extension officers cannot identify struggling learners, monitor which competencies are being mastered, or target their limited field visits where they are most needed, so support typically arrives only after a farmer has already suffered stock losses. Third, the lack of any competition or collaborative ranking mechanism removes the peer motivation that group-based extension and farmer cooperatives naturally provide, leaving learners isolated once training moves online. Fourth, existing approaches provide no self-regulated learning feedback, meaning farmers cannot track their own progress through the practical skills a successful production cycle requires, which research consistently shows is a prerequisite for sustained, self-directed skill development.

1.3 Research Objectives

The study aims to achieve the following three objectives:

1. To implement a gamification engine that awards points, badges, and leaderboard rankings based on measurable learner activities within a structured fish-farming curriculum, including lesson completion, quiz performance, and verified field-practice tasks.
2. To develop a real-time performance analytics dashboard for trainers and extension officers displaying individual and cohort-level metrics, including module completion rates, quiz scores, engagement frequency, and competency-progression trends.
3. To evaluate the effect of gamification mechanics on learner engagement and completion of fish-farming training modules through pre- and post-deployment surveys and platform usage data analysis at the pilot site.

1.4 Research Questions

The study is guided by the following research questions:

1. How can the gamification engine award points, badges, and leaderboard rankings in a way that sustains learner motivation across a full training programme?
2. How can the analytics dashboard provide trainers with accurate individual and cohort-level performance metrics from learner activity and assessment data?
3. To what extent do the gamification mechanics improve learner engagement and module completion at the pilot site?

1.5 Scope of the Study

This study focuses on the design, development, and pilot evaluation of a web-based Gamified Learning Platform with Performance Analytics for fish-farming skills training at a single Zambian site. The training content is organised around the core competencies of a small-scale aquaculture production cycle: pond siting and construction, water-quality management, fingerling selection and stocking, feed and feeding management, fish health and disease management, harvesting and post-harvest handling, and basic record-keeping and market access. The platform scope includes learner registration and enrolment; delivery of these training modules through text, images, and

demonstration video; auto-graded quizzes and scenario-based challenges; a gamification engine implementing points, badges, and leaderboard rankings; verified field-practice tasks in which learners submit photographic evidence for trainer verification; a real-time trainer analytics dashboard; and a learner progress dashboard. Because the target users are predominantly in rural, low-connectivity areas, the platform is designed to be lightweight and to support offline access to downloaded modules with synchronisation when connectivity is available. The study evaluates the platform through pre- and post-deployment learner surveys and platform usage log analysis. The study does not cover multi-site deployment, native mobile application development, artificial-intelligence-based personalised learning paths, livestock or crop training content beyond fish farming, or live video streaming. All evaluation is limited to the pilot learner cohort at the single participating site during the study period.

1.6 Significance of the Study

Increased Learner Engagement: Gamification mechanics create a self-reinforcing motivation loop. Points reward effort, badges recognise mastery of each farming competency, and leaderboards introduce healthy competition among farmers in the same cohort or cooperative. Together these sustain participation across a full training programme rather than the drop-off typical of one-off workshops.

Data-Driven Training and Extension: The real-time analytics dashboard enables trainers and extension officers to identify struggling farmers early, prioritise scarce field visits using cohort performance data, and direct support where it is most needed, transforming extension from reactive to proactive.

Visible Skill Achievement: The platform gives farmers tangible, immediate recognition of progress through visible XP totals, a collection of competency badges, and leaderboard standing. This turns an otherwise invisible learning process into a clear, motivating journey toward becoming a skilled fish farmer.

Bridging Learning and Practice: Verified field-practice tasks connect on-screen learning to the real pond. By submitting photographic evidence of a constructed pond, stocked fingerlings, or a feeding record for trainer verification, learners are rewarded for applying skills in practice, not merely for reading about them.

Addressing the Zambian Aquaculture Skills Gap: Zambia currently lacks locally developed, engaging digital training tools tailored to its small-scale fish farmers. This project directly addresses that gap with a locally built, low-bandwidth solution designed for the realities of rural farming.

Academic Contribution: The project provides empirical evidence on the effectiveness of gamification for agricultural skills training in a Zambian context, a significantly under-researched area, and produces a replicable platform architecture that can be extended to other agricultural sub-sectors.

1.7 Conceptual Framework

The proposed system operates through a four-component architecture that connects learner interactions to trainer insights via a central gamification engine. Figure 1.1 below illustrates this framework.

The Fish Farmer/Trainer actor interacts with the Training Platform UI, the React.js web interface through which farmers enrol, work through training modules, submit quizzes and field-practice tasks, and view their personal progress, and through which trainers upload content, configure assessments, verify field tasks, and access analytics. The Training Platform UI communicates bidirectionally with the Gamification Engine, the core processing system. The Gamification Engine evaluates every learner activity event, such as lesson completion, quiz submission, scenario challenge, verified field task, or login streak, against a configurable set of point-award and badge-unlock rules. It updates the learner XP total, checks badge criteria, and recalculates leaderboard rankings in real time. The Gamification Engine also communicates bidirectionally with the Trainer Analytics Dashboard, which aggregates learner activity data, calculates completion rates and competency-progression trends, and presents individual and cohort-level metrics to trainers. Below the Gamification Engine, the Points/Badge Module manages the specific award logic and badge inventory, communicating with the Learner Progress Store, the MySQL database layer that persistently records all learner XP, badge achievements, quiz scores, verified tasks, and activity history. This four-component design ensures that gamification incentives, analytics visibility, and persistent data storage are tightly integrated rather than added on as separate concerns.

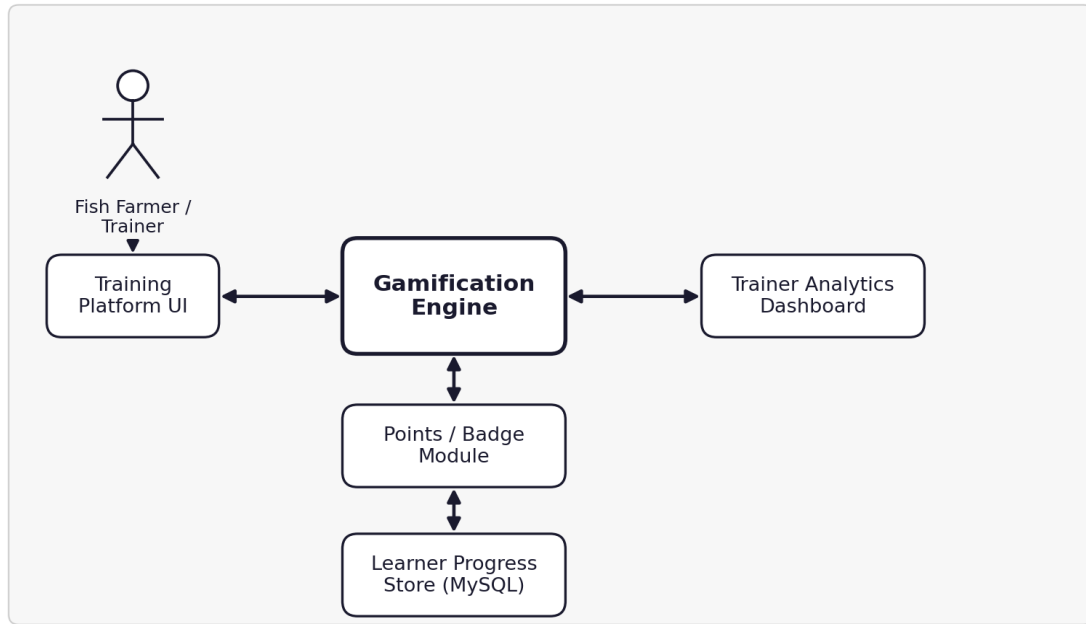


Figure 1.1: Conceptual Framework of the Gamified Learning Platform for Fish Farming Skills Training

1.8 Summary

This chapter established the rationale and research foundations for a Gamified Learning Platform with Performance Analytics for fish-farming skills training. It situated the project within Zambia's growing aquaculture sector and its persistent skills gap, identified the limited reach of conventional extension and the motivational and visibility deficiencies of current training approaches, articulated a problem statement covering four interconnected challenges, stated three measurable research objectives with corresponding research questions, scoped the study to a single pilot site with an offline-tolerant, competency-based curriculum, set out the system's significance across learner engagement, trainer insight, the link between learning and practice, and academic contribution, and described a four-component conceptual framework connecting learner interaction to gamification processing, analytics output, and persistent storage.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature and related works relevant to the proposed Gamified Learning Platform with Performance Analytics for fish-farming skills training. The review is structured around four thematic areas: the theoretical foundations of gamification, empirical evidence for gamification's effect on learning outcomes, learning analytics and trainer dashboard systems, and agricultural extension and aquaculture skills training in Zambia and Sub-Saharan Africa. For each theme, key works are examined, their contributions identified, and the gaps they leave clearly articulated to establish the specific contribution of the proposed system.

2.2 Related Works

2.2.1 Theoretical Foundations of Gamification

The authors in [1] published the seminal conceptual framework that formally defined gamification as the use of game-design elements in non-game contexts, and established a taxonomy of game-design components, including points, badges, leaderboards, progress bars, and narratives, that could be applied to digital products. Their framework distinguished between game elements, the structural features such as points and leaderboards, and game mechanics, the rule systems governing how elements interact. This distinction directly informs the design of the proposed platform's gamification engine.

However, the framework in [1] was purely conceptual and did not provide empirical evidence for the effectiveness of specific gamification elements, nor did it address the integration of gamification with trainer analytics or with practical, skills-based learning. This gap motivates both the empirical evaluation component and the analytics dashboard of the current project.

The authors in [2] extended this conceptual foundation by proposing a structured gamification design framework that organises game thinking into dynamics, mechanics, and components, and by setting out a six-step design process for applying these elements to real products. Their work provides practical guidance on selecting and combining points, badges, and leaderboards, the same component set adopted by the proposed platform. However, their framework was directed primarily at business and marketing applications, offered no treatment of vocational skills training,

and did not consider how gamification data could be surfaced to instructors, leaving the trainer-visibility problem unaddressed.

The motivational basis of gamification is most commonly explained through Self-Determination Theory, developed by the authors in [3], which holds that motivation is strongest and most durable when an activity satisfies the basic psychological needs of competence, autonomy, and relatedness. Building on this theory, the authors in [4] conducted a randomised experimental study isolating the effects of specific game-design elements, finding that badges, leaderboards, and performance graphs significantly increased perceived competence, while avatars, narratives, and teammates strengthened social relatedness. These findings justify the proposed platform's combination of competence-oriented mechanics, such as badges and visible progress indicators, with the socially oriented leaderboard. However, both works were conducted in laboratory and workplace settings in high-income countries and did not examine agricultural learners or low-connectivity environments.

The author in [5] developed a theory of gamified learning that formally distinguishes gamification from serious games and proposes that game elements affect learning outcomes indirectly, by changing learner behaviours and attitudes such as participation, effort, and persistence, which in turn influence learning. This theory informs the current project's decision to award points only for measurable, learning-relevant activities, including lesson completion, quiz performance, and verified field tasks, rather than for superficial interactions. However, the theory was presented largely as a research agenda and has not been validated in vocational or agricultural training contexts.

2.2.2 Empirical Evidence for Gamification in Learning

The authors in [6] conducted a systematic literature review of 24 empirical gamification studies published between 2010 and 2014, finding positive effects on engagement in the majority of studies, with motivational outcomes most consistently improved when gamification included social elements such as leaderboards alongside individual progress tracking. Their meta-analysis remains the most widely cited empirical foundation for gamification in digital contexts.

However, the authors in [6] noted that most studies evaluated gamification over short periods, typically four to eight weeks, with small samples, and that few measured substantive learning

outcomes such as knowledge retention or skill acquisition, as opposed to engagement proxies such as time-on-platform. The current project addresses both limitations by evaluating over a full training programme and measuring quiz and competency performance alongside engagement metrics.

The authors in [7] conducted a systematic review specifically of gamification in educational settings, reviewing 34 studies and finding that while gamification consistently improved engagement, the magnitude of the effect varied substantially with how it was implemented. Studies combining points with immediate feedback and visible progress tracking consistently outperformed those using points alone. This finding directly informs the current project's decision to combine points, badges, leaderboards, scenario challenges, and a personal progress dashboard rather than any single mechanic in isolation.

The authors in [8] conducted a comprehensive survey of gamification in theory and action, synthesising the conceptual literature alongside 31 empirical studies. They found that results across domains were mixed and highly sensitive to context and implementation quality, and cautioned that gamification applied uniformly, without tailoring to the target users and setting, frequently fails to produce benefits. This finding motivates the current project's requirements-gathering interviews with farmers and trainers at the pilot site, ensuring that the selected mechanics fit the local context. However, none of the surveyed studies addressed agricultural skills training or learners in Sub-Saharan Africa.

More recently, the authors in [9] systematically reviewed gamified learning in higher education and found consistent evidence of improved student engagement, attitude, and academic performance, with points, badges, and leaderboards being the most frequently used and most effective combination of elements. However, their review was confined to formal university settings with reliable connectivity and institutional learning-management systems, conditions that differ markedly from the rural, low-bandwidth, adult-learner environment targeted by the current project.

2.2.3 Learning Analytics and Trainer Dashboard Systems

The author in [10] introduced the foundational theoretical framework for learning analytics, defining it as the measurement, collection, analysis, and reporting of data about learners and their

contexts for the purpose of understanding and optimising learning and the environments in which it occurs. His framework positioned the dashboard as the primary instrument through which analytics insight is translated into instructional action, establishing the conceptual basis for the analytics component of the current project.

The authors in [11] evaluated 15 learning analytics dashboard implementations across education contexts, finding that dashboards providing both individual-level and cohort-level metrics were significantly more actionable for instructors than those providing only aggregate statistics. This finding directly shaped the design of the proposed analytics dashboard, which presents both individual learner progress cards and cohort-level trend charts for trainers.

The authors in [12] conducted a systematic literature review of 55 learning dashboard studies, finding that most dashboards rely on logged learner-interaction data, that completion rates, time-on-task, and assessment scores are the most commonly presented indicators, and, critically, that very few dashboards had been evaluated with their intended users in authentic settings. This evaluation gap directly informs the current project's plan to assess the trainer dashboard with extension officers and trainers at the pilot site rather than in isolation.

Similarly, the authors in [13] reviewed 93 studies of student-facing learning analytics dashboards and educational recommender systems, finding that fewer than half reported any evaluation of effects on learner behaviour or achievement, and recommending that future systems pair learner-facing progress views with instructor-facing analytics drawn from the same data. The proposed platform follows this recommendation by providing both a learner progress dashboard and a trainer analytics dashboard built on a shared activity and assessment data layer.

2.2.4 Agricultural Extension and Aquaculture Skills Training in Zambia and Sub-Saharan Africa

Across Sub-Saharan Africa, a growing body of work examines the role of digital tools in agricultural extension, where conventional services struggle with high farmer-to-officer ratios, limited mobility, and infrequent farmer contact. An influential early review by the author in [14] examined mobile phone-based ICT initiatives for agricultural extension across developing countries, concluding that ICTs can substantially reduce the cost of delivering information to farmers, but that rigorous evidence on learning and adoption outcomes remained limited. The

authors in [15] reviewed the use of information and communication technologies (ICTs) for disseminating agricultural information and services to smallholder farmers in the region, concluding that mobile and web-based platforms offer substantial potential to widen the reach of extension, while highlighting persistent gaps in their development and deployment. In a broader scoping review of digital agriculture in Sub-Saharan Africa, the authors in [16] similarly found that digital extension services and mobile tools can extend advisory support beyond what face-to-face extension alone can achieve, but that adoption remains constrained by low connectivity, limited device access, and digital literacy barriers. Complementing these reviews, the authors in [17] assessed the experimental evidence on digital agricultural advice and found that mobile phone-based advisory services produce modest but highly cost-effective improvements in input adoption and yields, while noting that sustaining farmer engagement with digital services over time remains an open challenge, precisely the challenge that the gamification component of the current project is designed to address.

Within Zambia specifically, the aquaculture skills gap is well documented. The Ministry of Fisheries and Livestock, through its National Fisheries and Aquaculture Policy [18], identifies inadequate training and weak extension delivery as constraints on the productivity of emergent and small-scale fish farmers, and prioritises improved vocational training and extension services. The authors in [19] provided a sector-wide overview and evaluation of Zambian aquaculture, finding that small-scale producers dominate the sector numerically but achieve low yields, in large part because of limited access to technical knowledge, quality inputs, and extension support. The authors in [20] examined commercialisation and upgrading in the Zambian aquaculture value chain and similarly identified weak technical skills and limited access to training as barriers preventing smallholder farmers from moving into higher-value production. Programmes such as WorldFish's Aquaculture Technical, Vocational and Entrepreneurship Training project [21] confirm that smallholder fish farmers are thinly spread over large areas and underserved by existing extension, and that practical skills in pond and farm management remain a key constraint on the sector.

On the use of gamification in this context, the authors in [22] conducted a systematic review of gamification in agricultural extension, analysing 23 studies and finding that gamification consistently improves farmer motivation, involvement, knowledge retention, and behaviour change, and that low-cost, context-appropriate designs are most effective in developing regions.

However, their review and the wider literature reveal that few studies have evaluated gamification for sustaining engagement in agricultural skills training, and fewer still in a Zambian aquaculture context. The current project responds directly to this gap by combining gamification with performance analytics in a platform built specifically for Zambian fish-farming training.

2.3 Review of Related Works

The reviewed works establish three consistent findings that motivate the current project. First, gamification demonstrably improves engagement when implemented with combinations of social and individual mechanics rather than single elements in isolation [4], [6], [7], a finding that informs the multi-mechanic design of the proposed gamification engine. Second, learning analytics dashboards are most effective when they present both individual and cohort-level data, enabling trainers to identify struggling learners and target support [11], [12], [13], which shapes the dual-view design of the proposed dashboard. Third, the agricultural-extension and aquaculture literature confirms a persistent skills gap among small-scale fish farmers in Zambia and Sub-Saharan Africa [18], [19], [20], and the under-exploited potential of engaging digital delivery to address it [15], [17], [22], establishing the direct relevance of a gamified, analytics-driven training platform to the local context.

2.4 Proposed System

The proposed Gamified Learning Platform with Performance Analytics addresses the identified gaps by combining a multi-mechanic gamification engine, implementing points, badges, leaderboards, and scenario-based challenges informed by the framework in [1] and validated by the empirical evidence in [4] and [6], with a dual-view analytics dashboard providing both individual and cohort-level metrics as recommended in [11] and [13]. The platform is designed specifically for fish-farming skills training in the Zambian context, responding to the aquaculture skills gap documented in [18], [19], and [20], and to the engagement and connectivity challenges identified in the agricultural-extension literature [15], [16], [22]. It will be empirically evaluated over a full training programme to address the short-duration limitation identified in the meta-analysis in [6].

2.5 Summary

This chapter reviewed literature across four themes: gamification theory, empirical evidence for gamification in learning, learning analytics dashboard design, and agricultural extension and aquaculture skills training in Zambia and Sub-Saharan Africa. The review confirmed that multi-mechanic gamification and dual-view analytics dashboards represent best practice based on available evidence, and that gamification has been shown to improve engagement and knowledge retention in agricultural extension. It also established that no locally developed, gamified digital platform for fish-farming skills training currently addresses the specific engagement and visibility challenges identified in the Zambian context. The proposed system's design decisions are directly grounded in and responsive to the reviewed literature.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology for the design, development, and evaluation of the Gamified Learning Platform with Performance Analytics for fish-farming skills training. It covers the software development model adopted, the requirements specification process, functional and non-functional requirements, software design specifications including the chosen design tool and diagram, development tools, the research timeline, and the budget.

3.2 Software Development Methodology

The project adopts the Agile software development methodology with iterative sprint cycles. Agile was selected because the platform integrates multiple interdependent subsystems, namely the gamification engine, the auto-grading and scenario module, field-task verification, the analytics dashboard, and the offline-synchronisation data layer, that benefit from incremental integration and testing rather than big-bang delivery. Each sprint targets a specific module, allowing feedback from farmers and trainers at the pilot site to be incorporated between sprints before the next module is built. Four sprint cycles are planned. Sprint 1 covers user authentication, enrolment, and training-content delivery with offline download. Sprint 2 covers quizzes, scenario-based challenges, auto-grading, and the points and badge engine. Sprint 3 covers field-task submission and verification, leaderboard ranking, the learner progress dashboard, and the trainer analytics dashboard. Sprint 4 covers offline synchronisation, system integration testing, performance optimisation for low-bandwidth conditions, user acceptance testing at the pilot site, and final documentation.

3.3 Requirements Specification

3.3.1 Requirements Gathering

Requirements were gathered through three methods. First, a systematic literature review of gamification implementations, learning analytics dashboards, and digital agricultural-training tools provided technical and functional requirements. Second, structured interviews were conducted with extension officers and trainers and with small-scale fish farmers at the pilot site to

identify the platform capabilities most valued by the target users, including preferred gamification mechanics, the most important competencies and analytics metrics, device and connectivity constraints, and acceptable response-time thresholds. Third, a review of existing learning and agricultural-training platforms, such as Moodle and mobile agricultural-advisory applications, was conducted to identify common features to replicate and the specific gaps in gamification, offline support, and analytics to address.

3.4 System Requirements

3.4.1 Functional Requirements

The following functional requirements were identified for the proposed system:

1. The system shall allow farmers to register, log in, and enrol in a fish-farming training programme through a web browser without requiring application installation.
2. The system shall deliver training content, including text, images, and demonstration videos, organised into competency-based modules covering the aquaculture production cycle.
3. The system shall allow learners to download modules for offline study and synchronise their progress when connectivity is restored.
4. The system shall provide auto-graded multiple-choice quizzes and scenario-based challenges with immediate feedback displayed to the learner upon submission.
5. The system shall allow learners to submit field-practice tasks with photographic evidence, and allow trainers to review and verify these submissions.
6. The gamification engine shall award configurable point values for every tracked learner activity, including content completion, quiz performance, scenario challenges, verified field tasks, and login streaks.
7. The gamification engine shall evaluate badge unlock criteria after every scoring event and award competency badges to learners who meet defined thresholds.
8. The system shall maintain a real-time leaderboard ranking all enrolled learners within a cohort by total XP, refreshed after every scoring event.

9. The learner dashboard shall display the learner's current XP total, earned badge collection, leaderboard rank, module completion percentage, and quiz score history.
10. The trainer analytics dashboard shall display individual learner progress cards showing completion rate, quiz average, total XP, verified-task count, and last active timestamp.
11. The trainer dashboard shall display cohort-level charts showing average quiz scores over time, module completion distribution, and XP progression.
12. The system shall alert trainers when a learner's quiz score falls below a configurable threshold or when a learner has been inactive beyond a configurable period.

3.4.2 Non-Functional Requirements

The following non-functional requirements were identified:

1. Performance: Quiz auto-grading results shall be displayed within 500 milliseconds of submission. Dashboard data shall load within 3 seconds on a standard 3G connection.
2. Offline capability and low bandwidth: Downloaded training modules shall be fully usable without an active connection, and the platform shall remain functional on 2G and 3G networks, with synchronisation occurring when connectivity becomes available.
3. Scalability: The database schema and API architecture shall support a minimum of 500 concurrent learners without degradation of response time.
4. Usability: The platform shall be fully functional on mobile browsers at 360px viewport width without horizontal scrolling, given that most target users will access it on entry-level smartphones.
5. Security: All user passwords shall be hashed using bcrypt. Session management shall use JWT tokens with a 24-hour expiry.
6. Availability: The platform shall target 99% uptime during the evaluation period at the pilot site.

3.5 Software Design Specifications

3.5.1 Design Tool

The primary design tool selected for the proposed system is the Unified Modelling Language (UML) Use Case Diagram. It was selected because the platform serves three distinct actor types, the Learner (a fish farmer), the Trainer or Extension Officer, and the System Administrator, each with clearly differentiated interactions with the system's subsystems. The Use Case Diagram captures these actor and system interaction boundaries in a format that is directly readable by non-technical stakeholders, including the trainers and site administrators at the pilot location, enabling effective requirements validation before development commences.

3.5.2 Design Diagram

The system will be modelled using a UML Use Case Diagram illustrating all actors and their interactions with the platform's core use cases: Enrol in Programme, Download Module, Complete Lesson, Submit Quiz, Attempt Scenario Challenge, Submit Field Task, View Personal Dashboard, View Leaderboard, Upload Training Content, Configure Assessment, Verify Field Task, View Analytics Dashboard, Configure Badge Rules, and Manage Users. The diagram will clearly show which use cases are accessible to Learners only, Trainers only, or both, and will identify the extend and include relationships between related use cases such as Submit Quiz and Receive Points Award.

3.6 Software Development Specifications

3.6.1 Software Development Tools

The following development tools and technologies are employed in the proposed system:

1. React.js [23]: The frontend JavaScript framework for building the learner and trainer web interfaces, providing a component-based architecture for the real-time updating of gamification scores and dashboard charts.
2. Progressive Web App (PWA) with service workers: Enables offline access to downloaded modules and background synchronisation, directly addressing the low-connectivity context of rural fish farmers.
3. Node.js [24] with Express: The backend JavaScript runtime and web framework, providing REST API endpoints for enrolment, content delivery, quiz and scenario scoring, field-task verification, gamification scoring, and analytics data aggregation.

4. MySQL [26]: The relational database management system for storing learner profiles, module content metadata, quiz and scenario items, XP records, badge achievements, verified-task records, and activity logs in a normalised schema.
5. Chart.js [25]: The JavaScript charting library integrated into the React.js dashboard for rendering responsive line charts and bar charts of learner performance data.
6. JWT (JSON Web Tokens): The authentication token standard used for secure session management across the React.js frontend and Node.js backend.
7. bcrypt: The password hashing library used to securely store learner and trainer credentials in the MySQL database.
8. Jest: The JavaScript testing framework used for unit testing gamification scoring logic, badge evaluation rules, and API endpoint behaviour.
9. Git and GitHub: The version control system and remote repository used for collaborative development and sprint-level change tracking.

3.7 Research Timeline

Table 3.1 below presents the project timeline broken down into weekly activities spanning April to November 2026, covering a total of 32 weeks.

Activity	Phase	Start Week	End Week	Duration
Proposal Writing	Planning	Week 1	Week 4	4 weeks
Literature Review	Planning	Week 1	Week 8	8 weeks
Requirements Gathering	Analysis	Week 5	Week 10	6 weeks
System Design (UML)	Design	Week 9	Week 14	6 weeks
Training and Gamification Module Dev	Build	Week 13	Week 22	10 weeks
Analytics Dashboard Dev	Build	Week 17	Week 26	10 weeks
Testing and Evaluation	Testing	Week 24	Week 29	6 weeks
Documentation and Report	Delivery	Week 28	Week 32	5 weeks

Table 3.1: Research Timeline for the Gamified Learning Platform for Fish Farming Skills Training

3.8 Research Budget

Table 3.2 below presents the estimated research budget for the project.

#	Item / Description	Unit	Estimated Cost (ZMK)
1	VPS Cloud Hosting for Node.js API and React.js frontend (6 months)	6 mo	K 1,260
2	Domain Registration (.com, 2 years) and SSL Certificate	1	K 780
3	MySQL Managed Database (cloud-hosted, 6 months)	6 mo	K 840
4	User Testing Incentives for pilot fish-farmer cohort (20 at K50 airtime/data)	20	K 1,000
5	Transportation for pilot site, cooperative and farm visits	Lump	K 1,500
6	Research Materials: IEEE Xplore access, aquaculture and EdTech references	1	K 900
7	Printing of proposal, interim and final report copies	9	K 480
8	Contingency	1	K 240
TOTAL			K 7,000

Table 3.2: Research Budget for the Gamified Learning Platform for Fish Farming Skills Training

3.9 Summary

This chapter described the Agile methodology adopted for the project across four sprint cycles, specified functional and non-functional requirements, including offline capability and low-bandwidth operation, gathered through literature review, trainer and farmer interviews, and platform analysis. It defined the UML Use Case Diagram as the primary design tool, documented the React.js, Node.js, MySQL, Chart.js, and PWA development stack, and presented an eight-activity weekly project timeline and an itemised budget of K 7,000. These specifications provide a complete blueprint for the development and evaluation of the Gamified Learning Platform for fish-farming skills training.

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